

# Shangmin Chen

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## EDUCATION

### Boston University

Bachelor of Arts in Computer Science

Boston, MA

September 2022 – May 2026

## EXPERIENCE

### Software Engineer

September 2024 – December 2024

BU Spark!

Boston, MA

- Developed full-stack food waste platform using Django, JavaScript, and SQLite, connecting 500+ students and reducing campus waste by 10%
- Built RESTful API with sub-100ms query times and deployed via CI/CD pipelines on Heroku
- Implemented Auth0 authentication and Django security middleware, ensuring zero security incidents
- Accelerated release by 1 week using Agile methodology in team of 5, contributing to 7 iterations

### IT Consultant

January 2024 – July 2025

Boston University Engineering IT

Boston, MA

- Automated software deployment for 100+ devices using Microsoft Deployment Toolkit and created custom USB deployment solution for Windows 11, reducing deployment time by 40%
- Managed 200+ user accounts and permissions via NAS, conducting access audits and boosting system uptime by 15%
- Enhanced endpoint security with CrowdStrike for 50+ devices, resolving 20+ tickets/week with 100% incident resolution rate

## PROJECTS

### Real-Time Analytics Pipeline | *Apache Kafka, Apache Flink, InfluxDB, React Dashboard, Docker Compose* [Month] 2026

- Built event-driven analytics system processing [X]K events/minute through Kafka topics with Flink stream processing, implementing windowed aggregations and complex event pattern detection (producer load testing)
- Designed time-series data storage in InfluxDB with optimized retention policies and continuous queries, supporting [X]-day historical analysis with [X]ms query response times (benchmark with InfluxDB CLI)
- Created real-time React dashboard with WebSocket streaming and D3.js visualizations, displaying live metrics with [X]ms end-to-end latency from event ingestion to UI update (browser performance profiling)

### Fitness Tracking Mobile App | *React Native, TypeScript, Expo*

August 2025

- Built cross-platform app with drag-and-drop workout creation using React Native Reanimated v3 and PanGestureHandler for exercise reordering
- Implemented persistent data storage with AsyncStorage and custom React hooks for state management across workout sessions
- Developed modular component architecture with search/filter functionality and modal-based exercise selection interface

### Crime Analytics & Forecasting Platform | *Python, Streamlit, Prophet, Scikit-learn, GeoPandas* December 2024

- Built crime analytics platform processing 500K+ Boston crime records with Prophet time-series forecasting and DBSCAN clustering for hotspot identification across 12+ police districts
- Implemented advanced data pipeline with seasonal decomposition and confidence intervals, generating 2-month crime predictions with train/test validation
- Developed interactive dashboard using Streamlit and Plotly, creating 121 crime-type heat maps with Folium and real-time geospatial visualizations

## TECHNICAL SKILLS

**Frontend Languages:** JavaScript, TypeScript, HTML/CSS, Python, SQL (PostgreSQL)

**Frontend Frameworks & Libraries:** React, React Native, Expo, Reanimated, PanGestureHandler, D3.js, Plotly

**Data Visualization:** D3.js, Plotly, Folium, Geospatial Visualizations, Real-Time Dashboards, Interactive Charts

**UI/UX & Animation:** React Native Reanimated v3, Gesture Recognition, Drag-and-Drop Interfaces, Modal Components

**Real-Time Features:** WebSocket Streaming, Real-Time Data Updates, Live Metrics Display, Performance Profiling

**Tools & Platforms:** Docker, Docker Compose, Git, Browser Optimization, Cross-Platform Development